

Quality Control Standard Operating Procedure

ForgeCore Industries — Quality Assurance Department

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1. Purpose and Scope

This SOP defines the quality control procedures for all crucibles manufactured at ForgeCore Industries' Pittsburgh facility. It covers incoming raw material inspection, in-process controls, finished-goods testing, and non-conformance handling. This procedure applies to all FC-series products.

2. Incoming Material Inspection

All raw materials must pass inspection before release to production:

Silicon carbide grain: certified purity $\geq 98.5\%$ SiC, $\text{Fe}_2\text{O}_3 \leq 0.6\%$

Graphite flake: certified carbon content $\geq 99.5\%$, ash $\leq 0.5\%$

Zircon sand: certified $\text{ZrO}_2 \geq 65\%$, $\text{TiO}_2 \leq 0.15\%$

Clay binder: moisture content 8-12%, plasticity index ≥ 25

Sample size: 5 kg composite per shipment; tested per ASTM C323, C324, C325

Acceptance criteria: all test results within supplier specification AND within ForgeCore internal specification. Any out-of-specification result requires QA Manager approval before material release.

3. In-Process Quality Control

Production operators must perform the following checks at the indicated frequencies:

Check	Frequency	Method	Acceptance
Batch weight	Every mix	Calibrated scale $\pm 0.1\text{kg}$	$\pm 1\%$ of recipe
Moisture	Every mix	Moisture analyzer	5-9%
Mold pressure	Every press	Pressure gauge	35-45 MPa
Green density	Every 10th piece	Dimensional + weight	$\pm 3\%$ of nominal
Kiln temperature	Continuous	Thermocouple + logger	Per firing curve $\pm 10^\circ\text{C}$
Firing duration	Continuous	Timer + kiln log	Per firing curve $\pm 30\text{ min}$

4. Finished Goods Testing

Sampling plan: ANSI/ASQ Z1.4, level II inspection, AQL 1.5 for major defects, 2.5 for minor.

Required tests per production lot (minimum):

Visual inspection: 100% of finished pieces

Dimensional check: 10 pieces per lot

Bulk density: 3 pieces per lot (ASTM C134)

Cold crushing strength: 1 piece per lot (ASTM C133)

Modulus of rupture: 1 piece per lot (ASTM C133)

Thermal cycling test: 1 piece per lot, 5 cycles to rated temperature

5. Non-Conformance Handling

Any product failing finished goods testing must be placed on QC hold with a red tag indicating lot number, defect type, and date. The QA Manager must disposition within 48 hours. Disposition options:

Rework: minor defects correctable within standard process

Relabel: downgraded to lower-tier product if functional

Scrap: dispose per environmental SOP ENV-002

Return to supplier: if defect traced to incoming material

6. Records and Retention

All QC records are stored digitally in the ForgeCore QMS (qms.forgecore.example.com). Records are retained for 7 years. COAs are provided to customers with each shipment and archived in the customer portal.